

ADVANCE MYSQL FOR INTERNET MOVIE DATABASE

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INTRODUCTION TO DATABASE MANAGEMENT SYSTEM(DBMS).

A Database Management System (DBMS) is like a digital notebook that helps you store, organize, and manage data — but smarter and much more powerful.

Feature What it Means

- Store data
- Keeps data in tables
- Retrieve data
- Lets you search or filter data
- Modify data
- You can add, update, delete records
- Security
- Controls who can access or change data
- Backup & Recovery
- Keeps your data safe from loss

RELATIONAL DATABASE MANAGEMENT SYSTEM

A relational database stores structured data in linked tables using rules and keys to keep it clean and connected.

Concept	Meaning
Table (Relation)	Holds data (like a spreadsheet)
Row (Record)	One entry (like one movie)
Column (Attribute)	One type of info (like title, year)
Primary Key	Uniquely identifies a row
Foreign Key	Links to a row in another table

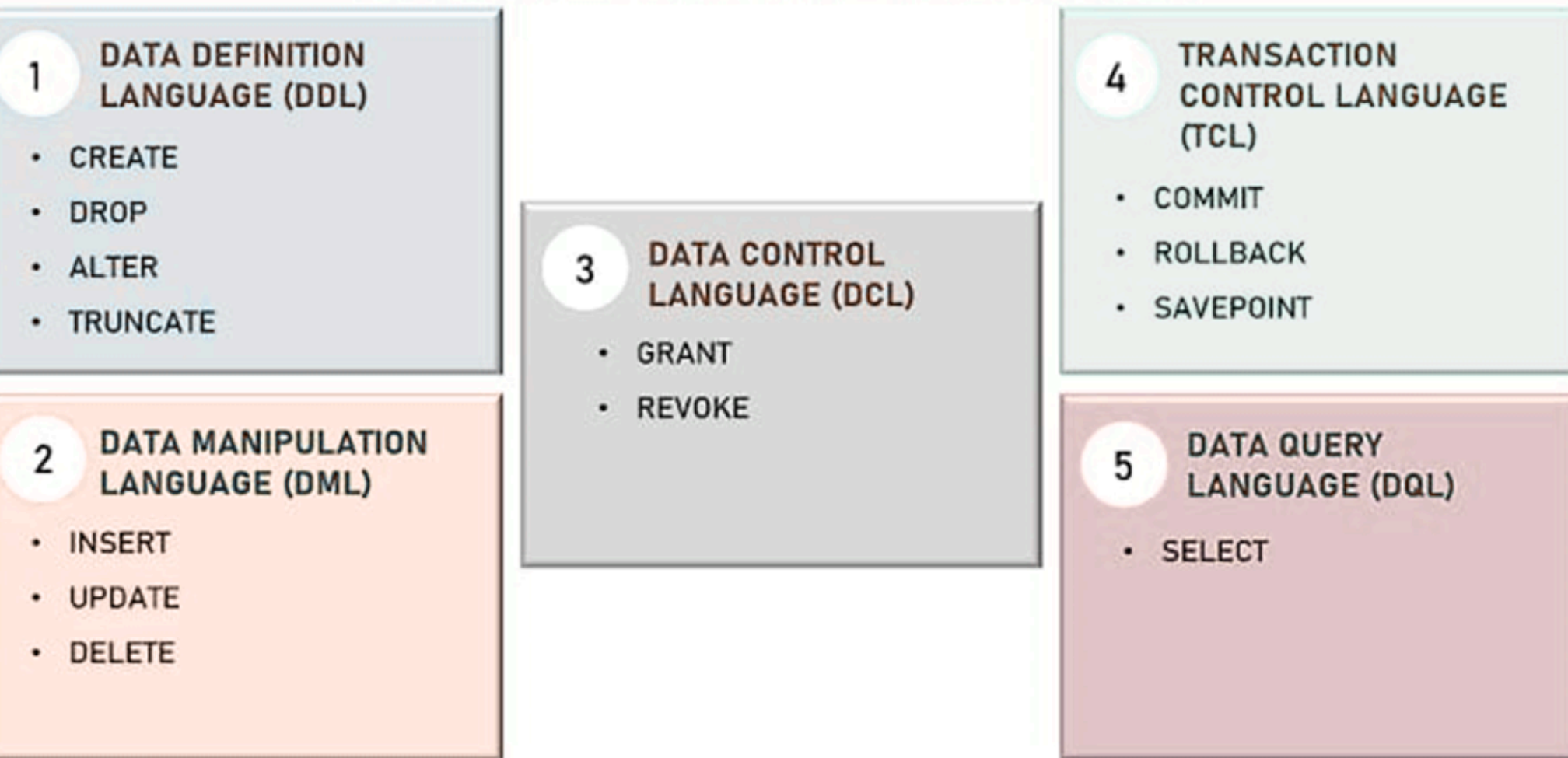
BENEFITS

1.Data consistency: Ensures data consistency across the database.

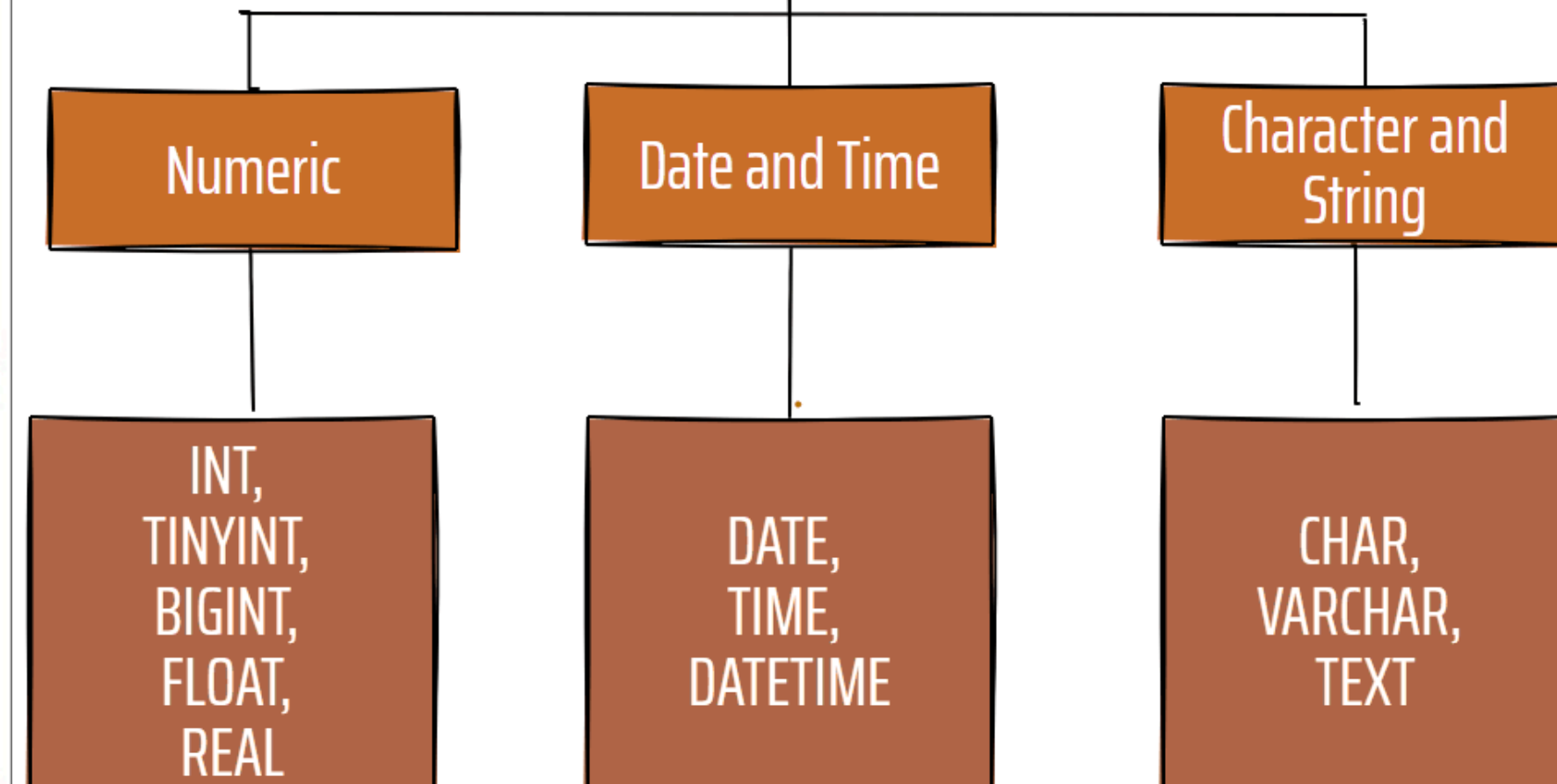
2.Data integrity: Supports constraints and relationships to maintain data integrity.

3.Query flexibility: Allows complex queries using SQL.

SQL COMMANDS



SQL Data Types



Summary Table

Join Type Includes from A Includes from B Missing side
shows as

INNER JOIN  Matching  Matching  Excluded

LEFT JOIN  All  If matching NULL

RIGHT JOIN  If matching  All NULL

FULL OUTER JOIN  All  All NULL

CROSS JOIN  All  All — Cartesian product

INTERNET MOVIE DATABASE

PROJECT OVERVIEW(IMDB)

Table Name	Description
movie	Stores basic details about each movie(ID,TITLE,DATE, PUBLISHED,COUNTRY,LANGUAGE,GROSS INCOME,PRODUCTION BUDGET)
genre	Lists genres for each movie (many-to-one).(MOVIE ID,GENRE)
director_mapping	Connects movies to directors using their IDs(MOVIE ID,NAME OF THE DIRECTOR)
role_mapping	Connects movies to people (actors, writers, etc.).(MOVIE ID,NAME ID OF ACTOR AND ACTRESS,CATEGORY)
names	Stores person details (actors, directors, etc.)(ID,NAME,DATE OF BIRTH,HEIGHT,KNOWN FOR)
ratings	Stores rating data for each movie.(MOVIE ID,AVERAGE RATINGS,TOTAL VOTES,MEDIAN RATINGS)

IMDB PRESENTATION USING SCREENSHOTS

HOW MANY MOVIES WERE PRODUCED IN EITHER THE USA OR INDIA IN THE YEAR 2019?



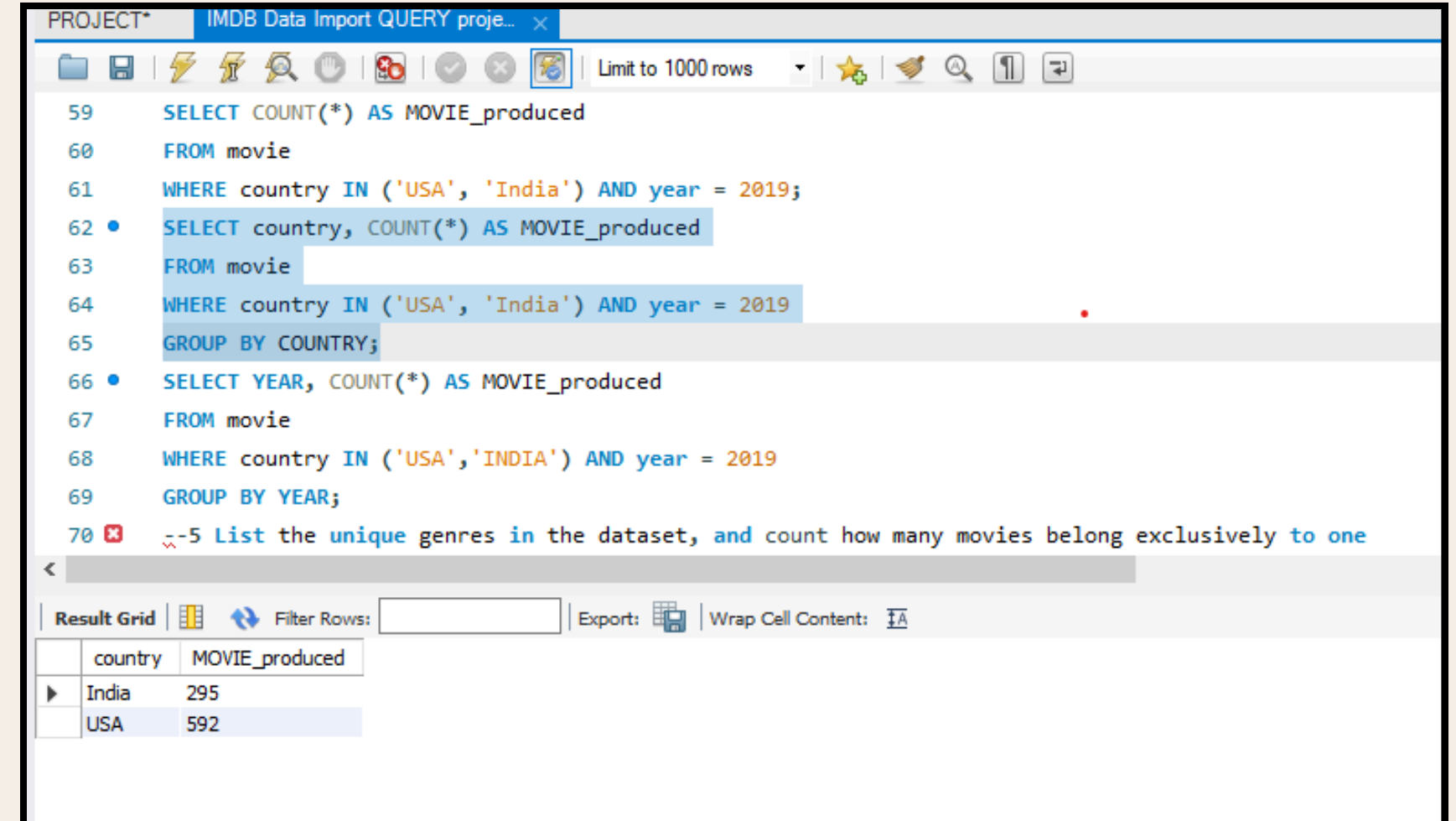
58 --4.How many movies were produced in either the USA or India in the year 2019?

```
59 SELECT COUNT(*) AS MOVIE_produced
60 FROM movie
61 WHERE country IN ('USA', 'India') AND year = 2019;
62 • SELECT country, COUNT(*) AS MOVIE_produced
63 FROM movie
64 WHERE country IN ('USA', 'India') AND year = 2019
65 GROUP BY COUNTRY;
66 • SELECT YEAR, COUNT(*) AS MOVIE_produced
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

MOVIE_produced
887

Result 19 x



PROJECT* IMDB Data Import QUERY proje... x

Limit to 1000 rows

```
59 SELECT COUNT(*) AS MOVIE_produced
60 FROM movie
61 WHERE country IN ('USA', 'India') AND year = 2019;
62 • SELECT country, COUNT(*) AS MOVIE_produced
63 FROM movie
64 WHERE country IN ('USA', 'India') AND year = 2019
65 GROUP BY COUNTRY;
66 • SELECT YEAR, COUNT(*) AS MOVIE_produced
67 FROM movie
68 WHERE country IN ('USA', 'INDIA') AND year = 2019
69 GROUP BY YEAR;
70 --5 List the unique genres in the dataset, and count how many movies belong exclusively to one
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

country	MOVIE_produced
India	295
USA	592

IMDB PRESENTATION USING SCREENSHOTS

6. WHICH GENRE HAS THE HIGHEST TOTAL NUMBER OF MOVIES PRODUCED?

The screenshot shows a database query editor interface. The query is as follows:

```
80 FROM genre
81 GROUP BY movie_id
82 HAVING COUNT(*) = 1
83 ) AS single_genre_movie;
84
85 --6. Which genre has the highest total number of movies produced?
86 SELECT genre, COUNT(*) AS total_movies
87 FROM genre
88 GROUP BY genre
89 ORDER BY total_movies DESC
90 LIMIT 1;
91
```

Below the query editor, the results are displayed in a table:

genre	total_movies
Drama	4285

The interface also includes a toolbar with various icons, a 'Limit to 1000 rows' dropdown, and a 'Result Grid' tab. The 'Result Grid' tab is active, showing the query results. The 'Output' tab is also visible at the bottom.

IMDB PRESENTATION USING

SCREENSHOTS

FIND THE MINIMUM AND MAXIMUM VALUES FOR EACH COLUMN IN THE RATINGS TABLE, EXCLUDING THE MOVIE_ID COLUMN.

PROJECT* IMDB Data Import QUERY proje... x

Limit to 1000 rows

```
104 HAVING COUNT(DISTINCT rm.movie_id) > 3;
105
106 ==9. Find the minimum and maximum values for each column in the ratings table, excluding the
107 movie_id column.
108 SELECT
109     MIN(avg_rating) AS min_avg_rating,
110     MAX(avg_rating) AS max_avg_rating,
111     MIN(total_votes) AS min_total_votes,
112     MAX(total_votes) AS max_total_votes,
113     MIN(median_rating) AS min_median_rating,
114     MAX(median_rating) AS max_median_rating
115 FROM ratings;
```

Result Grid

	min_avg_rating	max_avg_rating	min_total_votes	max_total_votes	min_median_rating	max_median_rating
▶	1.0	10.0	100	725138	1	10

Result 34 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 38	02:03:58	select * from ratings LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
✓ 39	02:04:34	SELECT MIN(avg_rating) AS min_avg_rating, MAX(avg_rating) AS max_avg_rating, MIN(...	1 row(s) returned	0.047 sec / 0.000 sec

IMDB PRESENTATION USING SCREENSHOTS

DETERMINE THE TOTAL NUMBER OF VOTES FOR EACH MOVIE RELEASED IN 2018.

The screenshot displays a database management interface with a menu bar (File, Edit, View, Query, Database, Server, Tools, Scripting, Help) and a toolbar. The left sidebar shows a 'SCHEMAS' tree with 'imdb' selected, containing 'Tables', 'Views', 'Stored Procedures', and 'Functions'. The main area shows a project named 'IMDB Data Import QUERY proje...' with a SQL query editor. The query is as follows:

```
--24. Determine the total number of votes for each movie released in 2018.
SELECT
  m.title,
  r.total_votes
FROM movie AS m
JOIN ratings AS r
  ON m.id = r.movie_id
WHERE m.year = 2018;
```

Below the query editor, the 'Result Grid' shows the following data:

	title	total_votes
▶	Le roi de coeur	3392
	The Other Side of the Wind	5014
	Kiss Daddy Goodbye	236
	The Evil Dead	174505
	Ek Din Achanak	179
	La Telenovela Errante	121

The bottom of the interface shows an 'Output' pane with 'Action Output' selected, displaying the following log:

#	Time	Action	Message	Duration / Fetch
✓ 2	18:52:38	USE IMDB	0 row(s) affected	0.016 sec
✓ 3	18:52:54	SELECT m.title, r.total_votes FROM movie AS m JOIN ratings AS r ON m.id = r.movie_id ...	1000 row(s) returned	0.031 sec / 0.015 sec

IMDB INSIGHTS

Insight Type	Example Query/Analysis
Top-rated movies	Find movies with highest avg_rating
Common languages in movies	Analyze the languages column
Prolific directors	Count number of movies directed by each person
Actor contributions	See which actors appear most frequently
Genre popularity	Count how many movies fall into each genre
Country-wise production	Number of movies produced per country





CONCLUSION

"To sum up, this project has helped me to achieve the main goal of the SQL concepts such as joins, aggregation, filtering, and grouping. I learned a lot through the process, and overcame challenges along the way. Believe this project can be useful for future use.

A photograph of a minimalist desk setup. A white desk is positioned against a light-colored wall. On the desk, there is a desk lamp with a blue shade and a brass base, which is turned on. A wooden chair is partially visible on the left side of the frame. A white rectangular box is overlaid on the center of the image, containing the text "ANY QUESTION?".

ANY QUESTION?



THANK YOU.

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@reallygreatsite